LADOKE AKINTOLA UNIVERSITY OF TECHNOLOGY, OGBOMOSO

COLLEGE OF HEALTH SCIENCES,

FACULTY OF CLINICAL SCIENCES, OSOGBO

DEPARTMENT OF MEDICINE

FINAL (PART IV) MBBS EXAMINATION (JUNE 5, 2018)

3:00 Hours

PAPER 2

INSTRUCTIONS:
ANSWER ALL QUESTIONS
EACH QUESTION SHOULD BE ANSWERED IN A SEPARATE BOOKLET
EACH QUESTION CARRIES 20 MARKS

1. A 22-year old male undergraduate complained of fever, cough and right-sided pleuritic chest pain all of 5 days duration
   a. Ask him further questions (3 marks)

On examination, he was found to be febrile (Temperature-38.8°C), dyspnoeic, and centrally cyanosed. He had a chest x-ray film from a Private Hospital, which showed consolidation of the right upper zone.

   b. What signs may you find in him? (1 mark)
   c. List five (5) relevant investigations (2 marks)
   d. What is your diagnosis? (1 mark)
   e. What is the most likely aetiological agent? (1 mark)
   f. Outline the management ($1^{1/2}$ marks)

2. A 40-year-old banker was rushed to the emergency ward of your hospital in the early hours of the morning. He was a known hypertensive of 5 years duration with poor drug compliance. He was said to have woken up with a severe headache and subsequently became confused and speaking irrationally. Blood pressure on admission was 190/120mmHg on admission.

   a. List four (4) differential diagnosis
   b. What is the most accurate diagnosis?
   c. How will you investigate this patient?
   d. How will you manage the index patient?
   e. List four (4) drugs indicated in the management of this patient.
   f. List four (4) complications of this condition.

3. A 55-year old female trader presents at Endocrinology clinic of LAUTECH Teaching Hospital with a 4-month history of tremor with intermittent anxiety and sweating. She has also lost some 12kg over this same period. He had intermittent increased defecation two months earlier. There is no neck swelling or pain and no palpitations. She has no visual disturbance. She has a history of hypertension, and she is a non-smoker. Mother had a

thyroidectomy before she died of breast cancer at the age of 85years. There is no history of recent domestic, financial or work stresses. She is on Lisinopril for hypertension.

a. List 3 possible diagnosis (differential diagnosis) in this patient - hyperthyroidism

Examination of the patient reveals "staring" gaze, fine tremor of out-stretched hand and palmar erythema.

b. What is the most likely diagnosis Grave's disease / hyperthyroidism.
c. Mention eight (8) possible cardiovascular signs that may be present in this patient
d. List four (4) relevant investigations and two (2) possible findings from each of the investigation ← TSH - radioactive uptake. T_3 - TSI T_4
e. List three (3) modalities of treatment ← medical, radiotherapy, surgical.
f. Outline three (3) common adverse effect of one of the modalities.

- tachycardia
- irregular pulse.
- wide PP.
- systolic HTN
- ↑ JVP
- hyperactive precordium
- S3 gallop
- murmur

4. A. A 35-year old female Nigerian was referred to the clinic because of increased skin pigmentation, progressive difficulty in straightening her fingers, digital ulcers, problem with mouth opening and paraesthesia of fingertips, and toes in the last two (2) years. She also had dysphagia, repeated history of heartburn, nausea, and vomiting with diarrhea since six (6) months before presentation. There was no history of joint pain or oliguria, and blood pressure on initial consultation was 90/60 mmHg.

a. What is your diagnosis? Systemic sclerosis.
b. Name four (4) possible cutaneous findings ← telangiectasia (1 mark) - pruritus. sclerodactyly (4 marks) - cutaneous ulcer. hyperpigmentation - thickened skin.

The patient returned to the emergency department three months after the initial consultation with sudden onset of breathlessness and reduction in urine output. Her blood pressure (BP) was 190/120mmHg, and the medical officer on call stabilized her.

c. What is your diagnosis at this point? complicated by renal insufficiency. (1 mark)
d. List four (4) relevant investigations to make this diagnosis (2 marks) ← renal USS - ESR. FBC. U,ICr - RF. Autoantibody assays.
e. List two (2) medications useful in the management with one example each (2 marks) - Prednisolone - Cyclosporin. - methotrexate - Rituximab. - Gold.

B. A 37year-old laboratory worker sustained needle injury while drawing blood from an Human Immunodeficiency Virus (HIV) positive client about two (2) years ago. She did not declare her injury or received any form of treatment. She presented to your clinic with a recurrent history of a headache, progressive weight loss, diarrhea, cough, and fever. Physical examination showed patient was pale, febrile, had oral thrush (Temperature 39°C), neck stiffness and kernig's was positive. A chest x-ray confirmed apical sparing, bilateral and diffuse opacity on the lungs fields. After counseling, the two antibody screening test using serial algorithm were positive for HIV.

a. She asked you to do a confirmatory test, what test would you order? Western blot analysis RT-PCR. (1 mark)
b. What WHO clinical stage is this patient? Stage 8 (1 mark)
c. List four (4) agents that might be responsible for her chronic diarrhea (2 marks)
d. List four (4) bacterial/ parasitic/ fungal agents that could be responsible for her neurological presentation diarrhea (2 marks)

- Shigella flexneri - Salmonella enteritidis. - Cryptosporidium parvum - Campylobacter jejuni: - CMV. - Adenovirus - Rotavirus:

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- Pneumocystis jiroveci
- Toxoplasma.
- mycobacterium avium complex.

She was investigated, treated and found eligible for treatment.

e. Write your prescription for a first-line highly active antiretroviral therapy and other opportunistic infections (OI) here present. (4 marks)

- Dapsone
- Fluconazole
- Dolutegravir
- Abacavir
- Lamivudine
- Clarithromycin
- Nystatin

5. A 33-year-old man presented with a bilateral flank mass of 3 years duration. There is a positive family history of a similar flank mass in the father who died of stroke six (6) years before patient's presentation. On examination, blood pressure (BP) was 162/106mmHg. Ultrasound shows multiple cysts in both kidneys.

a. What is the definitive diagnosis of this patient? (1 mark) PKD.
b. What are the genes responsible for this clinical condition? (2 marks) PKD-1 gene (Ch. 16) PKD-2 gene (Ch 4)
c. What are the proteins encoded by these genes? (2 marks) - Polycystin 1 & 2 [Ca-channel protein]
d. List four (four) other organs in this patient that cyst may be found (2 marks)
- liver
- pancreas
- spleen
- ovaries/testis
- large bowel
e. List three (3) other extra-renal manifestations of this condition apart from cysts in other organs and hypertension (3 marks)
- Saccular aneurysms
- valvular abnormalities
- Colonic diverticulae
- Abdominal wall & inguinal hernias
- Aortic root & annulus dilation

The patient developed sudden onset pain in the right flank

f. List four (4) likely causes of this pain (2 marks)
- Cyst rupture
- Nephrolithiasis
- UTI
- Large liver cysts

The estimated glomerular filtration rate of the patient is 72 ml/min, and current BP is 164/102nnHg, ECG shows left ventricular hypertrophy

g. How will you manage this patient taking into consideration likely diverse potential manifestations of he disease (8 marks)

Management
- Control HTN with ACEi or ARBs
- Control electrolyte imbalance e.g calcium
- Treat UTI with septrin and fluoroquinolones
- Treat hematuria with analgesics & oral hydration
- Pain relief analgesia. percutaneous cyst aspiration. sclerotherapy with alcohol.
- Reduce kidney fxn decline with tolvaptan [Vasopressin receptor V_2 antagonists]

MRA

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FACULTY OF CLINICAL SCIENCES

DEPARTMENT OF MEDICINE

90/5 = 18 mins

FINAL MBBS (PART IV) EXAMINATION IN MEDICINE

Paper 2: ESSAY

Date: 12th March, 2019

Duration: 2 hours

Answer all questions

1. A 35 year old male farmer was bitten on the left foot by a cobra (Elapidae family) 2 days ago while working on his farm. He now has hypoventilation and central cyanosis.
   a. Define clinical cyanosis
   b. Mention any 3 causes of hypoventilation
   c. What arterial gaseous pattern is expected in this condition?
   d. List 2 complications of snake bite
   e. What will be the difference in the amount of anti-snake venom (if there is envenomation) that would be administered to this 35 year old man and a 3 year old boy who has been bitten by same type of snake
   f. Where should this patient be managed and why?

alcohol, severe hypothyroidism
Drugs [narcotics]
Head trauma
chest wall deformity
Hypercapnia
Hypoxia
DIC - Rhabdomyolysis.
AKI - compartment syndrome.
- same dosage.
- ICU < intubation, multiparameter monitoring

2. A 24 year old 4th year undergraduate student presented with tiredness of one month duration. On examination, he was conscious, pale, afebrile and has mild ankle oedema. His pulse rate was 92/min, regular; blood pressure was 162/114 mmHg, with displaced heaving apex beat, and triple heart sounds. Urinalysis revealed 2+ proteinuria, and urine microscopy showed hyaline and red cell casts. The serum creatinine was 288 µmol/L, and estimated glomerular filtration rate is 25 ml/min/1.73m^2.
   a. List the likely diagnoses from the clinical features in this patient.

HF 2° CKD.

The haemoglobin concentration was 7.3g/dL.

   b. List six (6) relevant investigations to determine the cause of anaemia in this patient, stating the rationale for these investigations.
   c. List six (6) other relevant investigations in this patient and the possible findings (do not repeat investigations for anaemia and investigations already mentioned above).
   d. Outline the treatment of the patient giving appropriate rationale for such treatment

- Peripheral blood film.
- Abdpelvic USS
- Bone marrow studies
- Hb electrophor
- fecal occult test.
- Stool mcs.
- Serum vit B12; folate.

- transfuse at least 2 pints PCells or whole blood under lasix cover.
- treat

- FBC
- E, Cr, U.
- ECG
- ECHO
- LFT
- BNP
- Cardi
- GXM.
- Serology.

3. A 32 year old woman was admitted on account of muscle twitching, spasms, tingling sensation and numbness with associated short-lived self-limiting seizure prior to admission although she was alert on admission. The symptoms started since a thyroidectomy operation 4 weeks previously for a simple goiter. She has not attended endocrinology follow-up and was not taking any medication. There was no family or other personal history of note. Initial blood tests showed: Calcium-1.6mmol/l, Phosphate-1.8mmol/l, albumin-35g/L. Both vitamin D levels and alkaline phosphatase were normal.

35g/L = 350 g/dL (conv. mg/dL) / 18

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a. Calculate corrected calcium (2marks)
b. What is the definitive diagnosis? (2marks)
c. List 4 other causes of this condition (2marks)
d. List 2 signs that could be elicited in this patient and their interpretations (4marks)
e. How else can this diagnosis be confirmed and list 3 possible features that this investigative procedure would show (4marks)
f. List 4 hormones that regulate calcium metabolism and their actions (3marks)
g. Outline the principle of managing this condition (3marks)

4. A 52 years old executive administrator was rushed in from a business luncheon having suffered a severe chest pain which started one hour prior to that time and was described as very severe and referenced 10 on a scale of 10. He described the pain as tearing in nature and radiating to the back and to the jaw. Examination revealed a middle-aged man who was anxious, sweaty, and holding on to the chest. His blood pressure was 90/70 mmHg and pulse rate was 145 beats /min. He had a positive family history of sudden death and had been having previous history of chest pain which he adduced to his work schedule. Electrocardiography at the emergency unit revealed sinus tachycardia, left bundle branch block and left ventricular hypertrophy.
   a. List two differential diagnosis
   b. List 8 risk factors for this disease
   c. List 6 complications of this disease
   d. How will you investigate this patient?
   e. Highlight the principles of management of this clinical condition
   f. List 4 drugs that are indicated in the management of this condition

5. A 28-year-old female trader complained of progressively increasing raised rash involving the entire face, the nose, the ears and lip margins were not spared in the past 12 months. The upper and lower limbs were also involved. There was a positive history of paraesthesia of the lower limbs but no paralysis of any limb. About seven months after the onset of the rash on the limbs, some of the rash suddenly became painful, red and ulcerated. Examination of the skin showed symmetrically distributed erythematous and painful papules, nodules on the face with infiltration of the lip margins, ears, chin and the nose. There were erythematous painful ulcerated plaques, nodules and multiple fungating ulcers on the legs. There was no obvious limb deformity, or nerve thickening, no fever, lymphadenopathy, hoarseness or stridor, and no pedal edema. Her vision was clear and no eye pain or photophobia.
   a. List 5 common causes of ulcer in a medical ward 5 marks
   b. What is the most likely diagnosis? 2 marks
   c. Name precisely the phenomenon that occurred when the patient experienced sudden pain, redness and ulceration. 2 marks
   d. List three (3) relevant investigations. 3 marks
   e. List three (3) drugs of treatment and their dosages. 6 marks
   f. List two (2) long term complications of this condition. 2 marks

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FINAL MBBS (PART IV) EXAMINATION IN MEDICINE

Paper 2: Theory

Duration: 1 hour & 30 minutes **Date: 4th February, 2020**

1. A 42-year old man developed fever associated with chills and rigors, anorexia, and generalized body aches of 5 days duration. He presented at a Pharmacy outlet and was given artemeter/lumefantrine (LONART®) combination, and diclofenac potassium (CATAFLAM®) tablets. Fever, however, did not subside. He also took herbal medications which he purchased in the open market. He subsequently developed passage of watery stool barely 24 hours of ingesting the herbal medicine. He also noticed reduction in urinary output in the last 24 hours before presentation. At presentation at the Emergency Unit of the hospital, he is conscious, febrile (T-37.8°C), mildly pale, anicteric, acyanosed, dehydrated, no digital clubbing, no peripheral oedema. Respiration rate is 30 cycles/min, vesicular breath sound, with no crepitations or rhonchi. He has epigastric tenderness with no rebound tenderness, and liver, spleen and kidneys are not palpable. Pulse rate - 110/min, regular, normal volume, initial blood pressure (BP) at presentation was 92/56 mmHg, heart sounds were normal. Digital rectal examination was essentially normal. Rapid malaria test is negative.

   Full blood count shows Hb concentration of 9 g/dL, white cell count of 12,600/mm^3, neutrophils of 9,600/mm^3 with toxic granulations, platelet count of 120,000/mm^3. Urgent serum electrolyte and creatinine done shows Na^+ - 136 mmol/L, K^+ - 4.2 mmol/L, Cl^- - 98 mmol/L, HCO_3- 21 mmol/L, Urea – 9 mmol/L, serum creatinine – 160 µmol/L. Urinary output was 10 ml/hr. Repeat serum urea and creatinine shows urea of 20 mmol/L, and creatinine of 300 µmol/L.

   a. What are the possible differential diagnoses from these clinical presentations? [4 marks]

   b. List six (6) other relevant investigations that are required in this patient and the reason(s) for these investigations (do not mention investigations already done). [6 marks].

   c. Discuss the management of this patient. [10 marks]

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2. A 45 years old woman was brought to the emergency department of Ladoke Akintola University of Technology Teaching Hospital with three months history of progressive easy fatiguability, cough and paroxysmal nocturnal dyspnea. She was not a known hypertensive and her blood pressure on admission was 112/72mmHg. Examination revealed cardiomegaly, gallop heart sounds and hepatomegaly.

   a. List five possible aetiologies of the heart failure
   b. List six investigations that you will request for and one possible finding from each of the investigation mentioned.
   c. How will you treat this patient? Highlight the principle of management.
   d. List four drug classes that are of first-class indication in the condition
   e. List four complications of this condition.

3. A 48-year-old female patient had reported to the Dermatology Unit of the University Teaching Hospital with a chief complaint of recurrent raised itchy rash involving the trunk, the limbs and a burning sensation of the entire oral cavity in the past one (1) year. The patient gave a past medical history of being hypertensive and taking medications in the last 5 years. She takes usual Yoruba diet and she does not smoke or take alcohol. Examination of the skin showed symmetrical hyperpigmented papules and plaques involving the volar aspect of the trunk, the penis, the limbs including both wrists. The intraoral examination revealed a greyish brown patch with white striae.

   a. List four (4) **differential diagnoses** of this presentation
   b. List four (4) **variants** of the **MAIN** diagnosis
   c. What are the important three (3) **clinical signs** associated with the main diagnosis
   d. List four (4) **laboratory investigations** useful in the identification of metabolic complications of the main diagnostic entity.
   e. List three (3) **drugs** useful for treatment

4. A 75-year old man had worked in a stone cutting industry in the Northern part of Nigeria for 30 years before retiring 15 years ago and was placed on compensation by the company. He has developed constitutional symptoms of night fevers, weight loss, anorexia and respiratory symptoms of cough with sputum, chest pain, progressive breathlessness and lately moderate haemoptysis. He has grade 3 finger clubbing.

   a) What is the mostly likely diagnosis? (4marks)

   b) What is the mostly likely ventilatory pattern on spirometry (obstructive or restrictive pattern) State the expected measurement values or indices (2marks)

   c) What radiographic (CXR) pattern is likely in this patient? (2marks)

   d) Outline the management of this patient. (2marks)

5. A 35-year old man, a Nigerian, presented to the Emergency Department with a 4-week history of polyuria, polydipsia, and blurred vision. He started to vomit in the 24 hours leading to his presentation. She had no significant past medical history. His mother and paternal grandfather had type 2 diabetes mellitus. He smoked 20 sticks of cigarette per day and drank about 25 units of alcohol per week. Clinically he was dehydrated, tachypnoic, pulse rate 102/min, blood pressure 110/70mmHg. Physical examination revealed a waist circumference of 105cm, body mass index (BMI) of

31.7kg/m^2, no signs of cortisol excess, and was otherwise unremarkable. Causal blood sugar was 486mg/dl, and had significant (3+) ketonuria. Other laboratory results are bicarbonate 14mmol/l, urea 8.4mmol/l, creatinine 148micromol/l, sodium 138mEq/L, potassium 4.5mEq/l, chloride 103mEq/l and pH 7.28. A lipid profile shows the following, total cholesterol, 186mg/dl; triglycerides, 154mg/dl; LDL cholesterol, 112mg/dl; and HDL cholesterol, 47mg/dl.

(a.) List three (3) possible differentials in this patient [3mks]
(b.) Calculate anion gap, plasma osmolality & corrected sodium [3mks]
(c.) Identify as many cardiovascular risks present in this patient [4mks]
(d.) What is the definitive diagnosis [2mks]
(e.) List 4 other relevant investigations necessary in this patient with one reason each for the tests [4mks]
(f.) Enumerate the principles required in managing this patient [4mks]

RCC TB

Anion gap : [Na+K] - [Cl+Cl]

Osmolar gap :- 2(Na+K) + Glucose/18 + Urea/2.8

Corrected Na :- (Obs - Exp) x 0·3

	FEV$_1$	FVC	FEV$_1$/FVC
Normal	>80%	>80%	>80%
Rest	<80% pred	<80% p	<70% N
Obst	<80% pred	<80% p	<70% pred

Flattened diaphragm
Bullae formation
Hyperinflated lungs.

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DEPARTMENT OF MEDICINE
FINAL (PART IV) MBBS EXAMINATION (SEPTEMBER 2022)
PAPER 2

READ THE FOLLOWING INSTRUCTION CAREFULLY. Kindly answer all questions in this section in a different answer booklet.

DATE: 19th September 2022 TIME ALLOWED: 2 hours

1.

✓a. The management of type 2 diabetes goes beyond glycaemic control. Discuss this statement in NOT more than 2 pages (10 marks)

✓b. An 18-year-old female, Miss JD, is brought into the Accident and Emergency department from home by her father as she has been having abdominal pain, nausea and vomiting. She reports a loss of 5kg in weight over the preceding 4 weeks as well as feeling thirsty and reports passing urine frequently during both the day and night. Initial assessment notes that she has tachycardia (heart rate 134 beats per minute), blood pressure is 90/50mmHg and she is breathless with an increased respiratory effort and respiratory rate of 24 breaths/ minute. Her breath smells of nail varnish. The biochemistry laboratory result showed a blood glucose of 32 mmol/L (4-8 mmol/L) and bicarbonate of 7 mmol/L (21-28 mmol/L). Her urine dipstick is strongly positive for ketones (++++). She has no family history or past medical history of note. Miss JD is diagnosed as having diabetic ketoacidosis (DKA) as first presentation of Type 1 Diabetes Mellitus and is managed with intravenous fluids, insulin and potassium as guided by the LAUTECH TEACHING HOSPITAL, DKA protocol.

✓i. Give 3 symptoms in the history that are consistent with a diagnosis of Type 1 diabetes. (3 marks)

✓ii. What features of the examination and investigations lead the admitting team to decide that this presentation is in keeping with DKA? (3 marks – ½ mark for each up to 3)

✓iii. Briefly outline the pathophysiology of DKA (4 marks).

2. A 62 years old male retiree who is a known hypertensive of more than 15 years duration with questionable compliance was brought to the emergency unit of Ladoke Akintola University of Technology Teaching Hospital, Ogbomoso with complaints of difficulty in breathing at night associated with productive cough, chest pain. Oxygen saturation at the emergency unit on arrival was 82% while breathing ambient air. Examination revealed BP of 190/120 mmHg, gallop rhythm and bilateral oedema with the right leg more than that of the left one.

✓a. List 3 differential diagnoses (3 marks)

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b. List six investigations you will do for this patient with one abnormality that could be identified in the investigation. (6 marks)

c. Highlight the principles of management of this patient (6 marks)

d. List four drugs indicated to treat one of the differential diagnoses in this patient. (2 marks)

e. List 3 complications of one of the differential diagnosis (3 marks)

3. A 52-year-old woman, a known patient with systemic hypertension, was diagnosed 10 years ago but has not been regularly on follow-up and medication. The patient was brought to the Accident and Emergency Unit by her children following a day's history of headaches, blurring of vision, and reduction in urinary output (day-to-night ratio has reduced from 3:1 to 1:1). On examination, the patient is conscious, blood pressure on arrival was 230/120mmHg, and the apex beat is located in the 6^{th} left intercostal space, mid-clavicular line. Fundoscopy revealed optic disc swelling along with cotton wool spots, flame-shaped haemorrhages and hard exudates. Using the information provided, answer the following questions:

   a. Give three (3) differential diagnoses from this patient's clinical presentation.
   b. List the disease-mediated organ damage that may occur in this patient.
   c. List six (6) relevant investigations and their likely findings in this patient.
   d. How should blood pressure be controlled in this particular patient?
   e. List three (3) drugs that can be used to control the blood pressure in this patient. In addition, state the mechanisms of action of these drugs.
   f. Enumerate the lifestyle measures that will be helpful in the management of this patient.

4. A 34-year-old woman who was 24 weeks pregnant developed sudden onset of sore eyes, mouth and genitalia associated with blisters on her skin. She was admitted to the hospital by the obstetricians who referred her to the on-call dermatologist. The patient was diagnosed with HIV at antenatal screening and had commenced ART-Combivir and Nevirapine-2weeks earlier. Over the last 24 hours, she has felt unwell with a fever, malaise, sore throat, gritty eyes, and painful stomatitis. Initially, she had a few blisters on her face and upper trunk but over a few hours, these spread to involve extensive areas of her trunk and limbs. Her temp. was 38.2°C, B.P 180/100mmHg and pulse rate 100beats/min, CD4 counts 229cells/ml, HIV viral load 74,200 copies/ml, C-reactive protein 147mg/ml, Hb 9.7g/L, WBC 11.15 x 10^9 Neutrophils 8.29 × 10^9, Lymphocytes 1.28 x 10^9. Skin biopsy was taken.

   A. What is your diagnosis
   B. List 3 drugs that could give similar presentation in this environment in different class
   C. Outline your management in not more than ONE PAGE
   D. What parameters will you consider to assess her prognosis

5. An 82-year-old man living alone in a bungalow came to the clinic complaining of feeling generally unwell for about the last 3-4 months and of losing about 9.5 kg (21

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lbs) in weight during this period. On further enquiry he said he had been having high grade fever for the last month. He also has a past history of angina and arthritis and was on medication. On examination, he did not look well. He was pyrexic and without lymphadenopathy. Bibasal crepitus on the lower zone was heard on chest auscultation. He had hepatosplenomegaly and clubbing. Investigations showed WBC 12,300 cell/ul (neutrophils 52%, lymphocytes 39%), Hb 9.1 g/dl, with all other routine investigations being normal. A chest radiograph showed 1-2 cm diameter shadows all over the lung field. The Mantoux test was negative. No bacteria grew in a sputum culture.

a. What is the most likely diagnosis in this patient?
b. LIST 3 differential diagnosis
c. What are the other relevant laboratory tests in the diagnosis of this patient
d. Mention important bedside clinical finding that is indicative of the diagnosis in this patient
e. Mentions the appropriate medication for the management of this patient
f. Mention 2 complications of this clinical condition.

6. A 17-year-old lady was rushed into the emergency by her mother on account of nausea, vomiting, and right hypochondriac pain. She subsequently developed an altered level of consciousness. Her mother said she had swallowed 20 tablets of paracetamol 2 hours prior presentation because she failed her Joint Admissions Matriculation Board (JAMB) examination. She has no previous history of mental illness or suicidal attempts. On examination she was drowsy, had asterixis, she was also icteric with no other stigmata of chronic liver disease

a) What is the likely diagnosis based on the clinical presentation of the patient?
b) List 3 classes of drugs that can give this type of presentation
c) List 3 investigations you would carry out for this patient and the likely abnormalities
d) List 4 complications of the diagnosis
e) List 3 management options for this diagnosis (One must be definitive)

7. (a). Write a short essay on informed consent (not more than $1^1/_2$ pages). (6 marks)
(b). What is medical malpractice? (2 marks)
(c). List the four (4) elements that must be established to prove a claim of negligence in court. (2 marks)

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DEPARTMENT OF MEDICINE
PART IV MBBS EXAMINATION-
PAPER 2- ESSAY QUESTIONS

DATE: 16TH OF APRIL, 2024

DURATION: 3 HOURS

QUESTION 1

A 45-year-old of the Oyo State Civil Service was treated for community-acquired pneumonia at a private health institution. He was given some antibiotics, which he took for only four days, and discontinued the medication on his own because he felt much improved. He did not go for any follow-up, as advised by the doctor taking care of him.
One week later, all the earlier symptoms reappeared, but now with confusion, restlessness and cyanosis. The sputum microscopy, culture/sensitivity, and blood glucose level were normal, and the HIV test was also negative.

a. What is the most likely diagnosis?
b. What is the likely organism responsible?
c. How will you confirm the diagnosis?
d. Which class of antibiotics would you like to use for this type of organism?
e. What is the minimum number of days that this type of patient should be treated for before stopping the antibiotics?

QUESTION 2

QUESTION 2A

A young woman attends her family doctor with complaints of recent polydipsia and excessive urination. Otherwise, she is feeling healthy, although she has lost some weight even though her appetite has increased. Her body mass index (BMI) is 22.8 kg/m^2, her blood pressure is normal, and she has no family history of diabetes.

i. Based on the symptoms above, what is the most likely diagnosis and why? (2 marks)
ii. List three (3) possible tests that would confirm her diagnosis and what measurement would be accepted. (3 marks)
iii. List two (2) other important tests that will help support the possible diagnosis. (2 marks)

QUESTION 2B

Thyroid function tests were performed on a 52-year-old woman, and the following results were obtained:

Serum TSH: 6.0 mU/L (0.4-4.2),
Serum free T4: 5 pmol/L (10-20)
Serum free T3: 1.4 pmol/L (2.5-6.2)

i. What is the interpretation of this result? (2 marks)
ii. List 3 possible differential (causes) of this condition (3 marks)
iii. List 6 possible cardiovascular signs that may be present in this patient based on her condition. (3 marks)

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QUESTION 2C

A 54-year-old woman was referred to the outpatient department with weight loss, increasing pigmentation, and a supine systolic blood pressure of 50 mmHg. Addison's disease was suspected.

i. Briefly describe the dynamic test and the interpretation to confirm the diagnosis. (3 marks)
ii. List four (4) possible biochemical/metabolic abnormalities that may be present in this patient (2 marks)

QUESTION 3

QUESTION 3A

A 45-year-old staff of industry in Oyo town presented at the skin clinic of LAUTECH Teaching Hospital Ogbomoso with a 4-year history of recurrent moderately itchy papules and plaques involving the buccal mucosal, trunk and upper as well as lower extremities. Examination of the soft palate revealed a white linear streak. He has been on definitive treatment commenced at the skin clinic a year ago. He, however, presented at the clinic with a 3-month history of the appearance of new lesions following the use of tab lisinopril for his newly diagnosed systemic hypertension. Answer the questions below:

i. List three differential diagnoses (3 marks)
ii. What is your diagnosis (2marks)
iii. Mention two important diagnostic investigations (2 marks)
iv. What is the unique histologic picture of your diagnosis (1 mark)
v. List two drugs of choice (2 marks)

QUESTION 3B

A 22-year-old sexually active undergraduate student of a Nigeria College of Education presented at the Skin Clinic with multiple skin-coloured fleshy and umbilicated rash at the Mon pubis, penile shaft and anal area. He confessed to being an MSM. Retroviral screening done the same day revealed that he was reactive to HIV 1 but non-reactive to HIV 2. Answer the following questions:

i. What is the diagnosis of the genital lesions (2 marks)
ii. List two (2) relevant investigations to managing this patient (2 marks)
iii. Mention three (3) ARV drugs of choice in treating this patient (3 marks)
iv. List three (3) modes of treating the genital lesions (3 marks)

QUESTION 4

QUESTION 4A

Miss Blessing Johnson is a 29-year-old marketing officer at an IT firm who has had a 13-month history of recurrent headaches. Her headaches last about 6 to 10 hours per episode and she has at least an episode for between 15 to 18 days in a month. The headaches are unilateral, mostly on the left side of her head, but occasionally, she experiences the headache on the right side simultaneously. The headaches prevent her from carrying out her usual occupational tasks, and she has also missed some days of work due to the headaches. She explains that she sees bright spots and flashes of light before the onset of the headaches and notices frequent yawning hours before the headaches start. She complains of nausea but has no fever or neck pain and no weakness in her limbs. She takes alcohol occasionally, and she has been managed for depression by the mental health unit and uses diclofenac when the headaches start.

i. What is the patient's most likely diagnosis? (1 mark)
ii. List three possible triggers/precipitants for her conditions (1½ marks)
iii. Give one essential investigation to confirm the diagnosis (1 mark)
iv. List 2 groups of drugs that can abort the headaches and 1 group of drugs that can prevent the headaches in this condition (1½ marks)

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v. With the aid of a table, list the differences between migraine headaches and cluster headaches (4 marks)

QUESTION 4B

A 32-year-old bricklayer was referred to the Medical Outpatient Department by a doctor following incidental discovery of elevated blood pressure (BP) and persistently elevated serum creatinine over the last four months. On examination, he is conscious, pale, well hydrated, and has no pedal oedema. His pulse rate is 82/min, regular, normal volume, BP is 168/110 mmHg, Apex beat – 5th left intercostal space, lateral to the midclavicular line, HS 1,2,3 heard. Other systems are essentially normal. His serum creatinine is 222 µmol/L, and the estimated glomerular filtration rate (eGFR) is 26 ml/min/1.73m^2. The PCV is 26%.

i. List two (2) diagnoses in this patient. (2 marks)
ii. List eight (8) relevant investigations and the likely findings in this patient (4 marks).
iii. Briefly discuss the management of this patient (4 marks)

QUESTION 5

A 45-year-old male carpenter presented with a 4-month history of progressive abdominal swelling and generalized body weakness. He has been unable to go to his workplace. There is no significant weight loss. He has a history of sharing sharps and multiple unprotected sexual exposures. Examination revealed a chronically ill-looking man with fluffy hair and parotid fullness, with pedal swelling. Abdominal examination was significant for a distended abdomen with visible anterior abdominal wall veins, shrunken liver, and ascites demonstrable by fluid thrill. Ascitic tap yielded straw-coloured fluid. HBsAg was positive, and the HBV viral load was 20,000 IU/ml. Abdominal ultrasound revealed a liver with a wavy outline and coarse parenchyma echotexture.

i. What is the most likely diagnosis for this patient? (1 mark)
ii. List three (3) other signs that may be present in this patient (3 marks)
iii. List five (5) investigations you would carry out on this patient and the likely findings (5 marks)

He was started on diuretics (frusemide, Spironolactone) and other medications. He subsequently had large-volume paracentesis done on account of tense ascites causing respiratory distress. Forty- eight hours later, he was noticed to be sleeping excessively and later became drowsy

iv. What is the likely diagnosis of this second presentation and the precipitant? (2 marks)
v. List 4 other precipitants of the second presentation. (4 marks)
vi. Outline the management of the patient. (5 marks)

QUESTION 6

A 27-year-old lady was brought to the emergency ward with a history of paroxysmal cough, easy fatiguability, and leg swelling for two months duration. She has had a history of recurrent sore throat in the past and skin rashes. There was also history of palpitations and chest pain. Examination revealed cardiomegaly, blood pressure was 100/60 mmHg, pulse rate was 96/min, and heart rate was 116 beats per minute. There were bibasal crepitations on chest examination and Kerley B lines on Chest X-ray. She also had a history of recent weakness of the left upper and lower limbs, which occurred suddenly while she was having her bath a few days ago.

a. What is the primary diagnosis? (1 mark)
b. List two complications that have occurred in this patient (2 marks)

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c. List six (6) relevant investigations that should be done on this patient and the likely findings. (6 marks)
d. List the principles of management of this patient (7 marks).
e. Discuss the preventive strategy for this disease condition (4 marks).

QUESTION 7

a. Write a short essay on interprofessional collaboration and its core competencies. (10 marks)
b. Write a short essay on good medical practice and the domains of good medical practice (10 marks)

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COLLEGE OF HEALTH SCIENCES
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF INTERNAL MEDICINE
MBBS PART IV EXAMINATION
PAPER 2
JANUARY 21, 2025

QUESTION 1 GI

A 50-year-old man, a known alcoholic of 25 years who still drinks actively, presented to the emergency with sudden onset difficulty in breathing about 6 hours before presentation, which progressively worsened. There was a preceding history of early satiety and abdominal pain. Examination revealed a chronically ill-looking man who was not pale and anicteric but had bilateral pitting pedal oedema up to mid-leg with asterixis. Abdominal examination showed a grossly distended abdomen with visible anterior abdominal wall veins, the liver span was 10 cm, and there was an enlarged spleen and ascites demonstrable by fluid thrill.

A. List 3 likely causes of the difficulty in breathing (3 marks)
B. What five (5) other complications of the primary diagnosis can he present with? (5 marks)

Samples were taken for ascitic fluid analysis and Liver function test was done with the result shown below; Bilirubin 25 µmol/l, Conjugated bilirubin 4µmol/l, Total protein 69 g/l, Albumin 28 g/l, ALT – 8 IU/l, AST – 20 IU/l, ALP – 78 IU/l, GGT – 45 IU/l, INR 1.7, Ascitic Protein- 45g/l, Ascitic Albumin- 15g/l, Ascitic Neutrophil 268 cells/cm3, Ascitic WBC- 550 cells/cm^3; PCV- 41%, WBC- 7,800cm3, Platelet -240,000cm3

C. What is the serum albumin-ascitic gradient? (3 marks)
D. What is the diagnosis based on the clinical history and ascitic fluid analysis? (5 marks)
E. What is the definitive management for the background condition, and name one major contraindication to the management that this patient has? (4 marks)

QUESTION 2A Derm

A 42-year-old female lawyer presents with some lesions on her shins. The lesions are patches of shiny, red-brown skin with a yellow centre: some early ulceration exists in one area. You perform a biopsy, which shows granuloma, among other histologic findings. After starting the patient on super-potent topical steroids, you organise for an overnight fasting glucose, which is reported as 9.2 mmol/L. The patient is upset as she read on the internet that all people with lesions on the shin have diabetes. Answer the following questions:

a. What is the name of the lesion on the shin confirmed by biopsy?	2 marks
b. List other four (4) cutaneous features of diabetes mellitus	2 marks
c. List four (4) cutaneous complications of super-potent steroid	2 marks
d. List two (2) approaches to prevent topical effects of super-potent steroids	2marks
e. List two (2) other methods of treating this patient's cutaneous condition	2 marks

QUESTION 2B Infectious

A 42-year-old polygamist presented at the Retroviral clinic following a report of a positive HIV type 1 result after a febrile illness about 3 weeks before presentation. He had a diffuse, itchy papular eruption 6 months earlier. Further examination showed that he had a whitish lesion on the lateral surface of his tongue, which bled in an attempt to scrape with his toothbrush. His viral load was 1200 cells/mm^3. His complete

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blood counts revealed anaemia, lymphocytosis and elevated C-reactive protein and ESR. Answer the questions below:

A. What is your diagnosis? 2 marks
B. What is the oral lesion in this patient 2 marks
C. List two (2) other useful investigations in this patient 2 marks
D. Mention two (2) useful ARV agents in this patient 2 marks
E. List two (2) useful agents to treat this patient's oral lesion 2 marks

QUESTION 3A Stroke

A 72-year-old man presents to the Emergency Room with a sudden-onset right-sided weakness and slurred speech. His symptoms started 1 hour ago while he was watching television. He has a history of hypertension and type 2 diabetes mellitus but is otherwise well. He has a significant history of cigarette smoking and alcohol ingestion. On examination, he is alert but has a right-sided facial droop, weakness in the right upper and lower limb (power 2/5 in the upper limb and 3/5 in the lower limb), and dysarthria. His blood pressure is 180/100mmHg, and his blood glucose is 8.2 mmol/L. A cranial computer tomography scan was done, and it revealed no haemorrhage.

I. What is the most likely diagnosis? (1 mark)
II. Outline the immediate management steps for this patient. (3 marks)
III. List two indications and four contraindications for thrombolysis in a patient like this. (3 marks)
IV. List 3 investigations that should be performed to determine the cause of this stroke (1.5 marks)
V. What are the key components of secondary prevention for this patient? (1.5 marks)

QUESTION 3B Renal

A 28-year-old food vendor presents at the medical out-patient clinic on account of a 5-day history of pain on micturition with associated frequency and urgency. She recently noticed pain in her lower abdomen. There is no history of loin pain, and she has not noticed any change in the appearance of her urine. On examination, she has a temperature of 37.1 degrees Celsius; she is not pale and has no body swelling. On abdominal examination, she has tenderness in the suprapubic region. All other findings were essentially normal.

I. What is the patient's likely diagnosis? (1 mark)
II. How will you confirm your diagnosis and give the findings required? (2 marks)
III. List 5 factors that can make this patient susceptible to this condition (2.5 marks)
IV. Outline the management of this patient (2.5 marks)
V. List 4 advice you will give this patient to prevent a reoccurrence of this condition. (2 marks)

QUESTION 4 Cardio

A 62-year-old male retiree who is a known hypertensive of more than 15 years duration with questionable drug compliance was brought to the emergency unit of LAUTECH Teaching Hospital, Ogbomoso, with complaints of difficulty in breathing at night associated with productive cough and chest pain. On arrival, oxygen saturation at the emergency unit was 82% while breathing ambient air. Examination revealed a BP of 190/120 mmHg, gallop rhythm, and bilateral pitting pedal oedema.

A. List 3 differential diagnoses.

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B. List six investigations you will do for this patient, with one abnormality that could be identified on the investigation.
C. Highlight the principles of management of this patient.
D. List four drugs indicated to treat one of the differential diagnoses in this patient.
E. List 3 complications of one of the differential diagnoses.

QUESTION 5 Vascular

A 42-year-old man presented with a 3-week history of left leg swelling and pain. Physical examination revealed a tender, swollen calf with associated differential warmth and pitting pedal oedema. A complete blood count showed the following:

- Packed cell volume: 28%
- White blood cell count: 12,200 cells/mm^3
- Platelet count: 130,000 cells/mm^3

A. List three (3) differential diagnoses (3 marks)
B. What is the most likely diagnosis? (2 marks)
C. List two compounds that can be assayed in the serum to support the most likely diagnosis. (4 marks)
D. List five (5) investigations and the reasons for these investigations that you will order to help you evaluate this patient's state of health (5 marks)
E. Discuss the management of this patient (6 marks)

QUESTION 6 Endo

A 43-year-old woman is brought to the hospital Emergency Department after collapsing on the train on the way to work. She says she stood up to get off the train and felt very dizzy and then lost consciousness, fully recovering a few moments later. She has felt dizzy, especially when standing up, several times over the previous weeks, during which time she has also been feeling very tired. She has lost weight over this period, and her periods have stopped. Her prior medical history includes vitiligo, and she does not take any medications.

On examination, she appears tanned (apart from some hypopigmented patches on her arms, which she explained are caused by vitiligo), though she has not been in the sun for many months. Areas of pigmentation are also noted on the buccal mucosa inside her mouth and in the creases of her palms. Her blood pressure is 102/70 mmHg supine, falling to 76/62 mmHg after standing for 3 minutes. Otherwise, the physical examination is unremarkable. Her chest X-ray is unremarkable.

A. List the differential diagnosis for someone with a history of collapse and postural hypertension. (5 marks)
B. What is the likely diagnosis in this case? (3 marks)
C. List four (4) possible biochemical abnormalities that may be present in this case. (4 marks)
D. Describe the further investigations to confirm this diagnosis. (5 marks)
E. Outline important management steps for this case (3 marks)

QUESTION 7

"The global cancer burden is growing rapidly and is expected to have a differentiated impact on countries, based on their ability not only to treat cancer but also to effectively prevent cancer". (International Agency for Research on Cancer).

Discuss the primary prevention of cancer. (20 marks)

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